Unused descriptors versus DFT targets: Pearson r
Common training cohort; pairwise complete observations

| | OH energy | O energy | OOH energy | DFT overpotential |
|---|---|---|---|---|
| Adsorption metal: mean ligand dipole magnitude | -0.24 | -0.20 | -0.25 | +0.16 |
| Adsorption metal: spread ligand dipole magnitude | +0.11 | +0.06 | +0.12 | -0.16 |
| Adsorption metal: minimum ligand dipole magnitude | -0.18 | -0.13 | -0.19 | +0.16 |
| Adsorption metal: maximum ligand dipole magnitude | -0.59 | -0.60 | -0.61 | +0.10 |
| Adsorption metal: total ligand charge | +0.23 | +0.18 | +0.24 | -0.14 |
| Adsorption metal: mean ligand charge | +0.23 | +0.18 | +0.24 | -0.14 |
| Adsorption metal: spread ligand charge | +0.14 | +0.09 | +0.15 | -0.16 |
| Adsorption metal: minimum ligand charge | +0.59 | +0.60 | +0.60 | -0.02 |
| Adsorption metal: maximum ligand charge | +0.18 | +0.14 | +0.20 | -0.15 |
| Adsorption metal: total ligand spin magnitude | -0.02 | -0.00 | -0.05 | +0.22 |
| Partner metal: mean ligand dipole magnitude | -0.17 | -0.13 | -0.19 | +0.14 |
| Partner metal: spread ligand dipole magnitude | +0.15 | +0.11 | +0.17 | -0.16 |
| Partner metal: minimum ligand dipole magnitude | -0.16 | -0.12 | -0.18 | +0.15 |
| Partner metal: maximum ligand dipole magnitude | -0.14 | -0.14 | -0.15 | -0.02 |
| Partner metal: net ligand dipole magnitude | +0.13 | +0.09 | +0.15 | -0.16 |
| Partner metal: total ligand charge | +0.18 | +0.14 | +0.20 | -0.14 |
| Partner metal: mean ligand charge | +0.18 | +0.14 | +0.20 | -0.14 |
| Partner metal: spread ligand charge | +0.17 | +0.12 | +0.18 | -0.15 |
| Partner metal: minimum ligand charge | +0.20 | +0.20 | +0.22 | -0.05 |
| Partner metal: maximum ligand charge | +0.18 | +0.13 | +0.19 | -0.14 |
| Partner metal: total ligand spin magnitude | -0.00 | +0.02 | -0.01 | -0.02 |
| Mean ligand angle at adsorption metal | -0.14 | -0.07 | -0.12 | +0.03 |
| Spread of ligand angle at adsorption metal | -0.13 | -0.07 | -0.12 | +0.04 |
| Smallest ligand angle at adsorption metal | -0.17 | -0.13 | -0.16 | -0.02 |
| Largest ligand angle at adsorption metal | -0.14 | -0.07 | -0.12 | +0.03 |
| Mean metal-bridge-metal angle | +0.22 | +0.19 | +0.20 | -0.02 |
| Spread of metal-bridge-metal angles | +0.20 | +0.19 | +0.12 | +0.13 |
| Mean bridge dipole magnitude asymmetry | -0.02 | -0.01 | -0.09 | +0.17 |
| Partner metal spin magnitude | -0.04 | -0.04 | -0.05 | -0.05 |
| Signed metal spin product | +0.35 | +0.37 | +0.37 | +0.09 |
| Net support spin | +0.22 | +0.20 | +0.13 | +0.11 |
| Total support spin magnitude | -0.29 | -0.26 | -0.28 | +0.02 |
| Ligand nonplanarity | NA | NA | NA | NA |
| Metal displacement from ligand plane | +0.14 | +0.14 | +0.16 | +0.08 |
| Shared ligand count | NA | NA | NA | NA |
| Terminal ligand count | NA | NA | NA | NA |
| Adsorption metal: mean shared ligand charge | +0.19 | +0.14 | +0.20 | -0.14 |
| Partner metal: mean shared ligand charge | +0.19 | +0.14 | +0.20 | -0.14 |
| Partner metal: mean terminal ligand charge | -0.02 | -0.01 | +0.04 | -0.04 |
| Second-shell carbon density near 2.8 Angstrom | +0.03 | +0.08 | +0.01 | +0.14 |
| Second-shell carbon density near 3.6 Angstrom | +0.02 | -0.03 | +0.04 | -0.15 |
| Second-shell N/O/S density near 2.8 Angstrom | -0.15 | -0.12 | -0.11 | -0.03 |
| Second-shell N/O/S density near 3.6 Angstrom | -0.18 | -0.16 | -0.14 | -0.03 |
| Mean ligand angle at partner metal | -0.18 | -0.14 | -0.18 | +0.13 |
| Spread of ligand angle at partner metal | -0.16 | -0.11 | -0.16 | +0.14 |
| Smallest ligand angle at partner metal | -0.24 | -0.21 | -0.24 | +0.12 |
| Largest ligand angle at partner metal | -0.18 | -0.14 | -0.18 | +0.13 |

